Online Resource 1

Supplementary figures and tables

Article: *[Article title]*
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Cell	*In Vivo* Firing Rate (Hz)	Reference	Model Firing Rate (Hz)	*In Vivo* Activation	Reference	Model Activation
Mature granule	1.2 (0.56–2.38)	McHugh et al. (2022)	1.08	2–5%	Galloni et al. (2022); Hainmueller and Bartos (2018)	8.0%
	0.49	Diamantaki et al. (2016)		9%	GoodSmith et al. (2017)	
	0.031 ± 0.096	Zhang et al. (2020)				
	0.4	Wheeler et al. (2023)				
	0.96 ± 0.19	GoodSmith et al. (2019)				
	≈ 0.2	Senzai and Gyorgy Buzsaki (2016)				
	72 ± 8^{max}	Lübke et al. (1998)				
Immature granule	2.3 (1.6–6.7)	McHugh et al. (2022)	2.96			63.8%
Mossy	0.84	Wheeler et al. (2023)	3.45	$\approx$ 100%	Danielson et al. (2017)	96.7%
	1.05 ± 0.07	GoodSmith et al. (2019)		88%	GoodSmith et al. (2017)	
	≈ 0.8	Senzai and Gyorgy Buzsaki (2016)				
	50 ± 6^{max}	Lübke et al. (1998)				
HIPP	—		6.38			100%
Basket	13.6 ± 0.8	Li et al. (2022)	17.35			100%
	230 ± 15^{max}	Lübke et al. (1998)				
CA3 pyramidal	0.2	Mizuseki and Buzsáki (2013)	1.17	29%	GoodSmith et al. (2017)	30.0%
	0.5	Kay et al. (2016)		20–30%	Galloni et al. (2022)	
	0.72 ± 0.51	Lasztóczi et al. (2011)				
	CA3a: 0.4; CA3b: 0.3	Oliva et al. (2016)				
	1.2	McHugh et al. (2022)				
	0.92 ± 0.06	GoodSmith et al. (2019)				
	≈ 0.6	Senzai and Gyorgy Buzsaki (2016)				
CA3 inhibitory	20 ± 7	Tukker et al. (2013)	6.51			65.4%
	17 ± 7	Lapray et al. (2012)				
	8.2 ± 5.6	Leão et al. (2012)				

Table S1: *In vivo* firing rates and activation (the fraction of active cells) for each cell type, alongside model values. *In vivo* firing rates are mean ± SD. ^{max} maximum observed rate. ≈ approximate value read from figure. Model firing rates are the mean over active cells and model activation the fraction of active cells; both are from the control network, except the immature granule cells (absent in the control), which are from the neurogenesis model at 50% EC→iGC excitability.

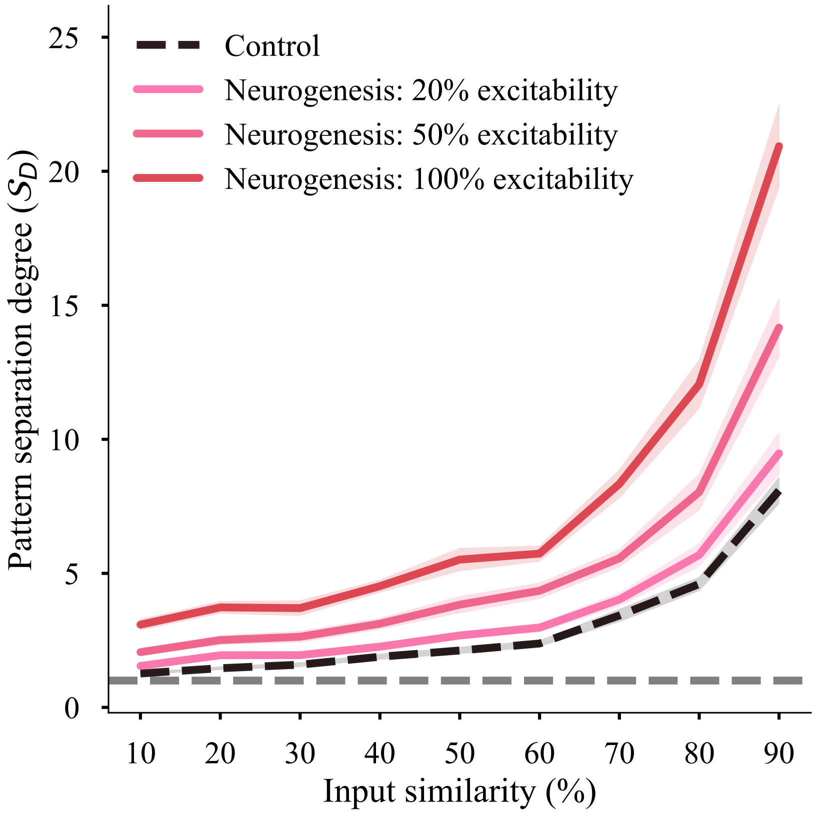

Figure S1: Pattern separation of the mature granule cells resolved by input similarity. Each curve shows the pattern separation degree $\mathcal{S}_D$ of the mGC population as a function of the similarity between the input patterns, for the control network without neurogenesis (black dashed line) and for neurogenesis networks at representative EC$\rightarrow$iGC excitability levels. Each point is the mean over the 30 pattern sets and the shaded band shows the standard error of the mean. The grey dashed line marks $\mathcal{S}_D = 1$, above which the output patterns are more distinct than their inputs. Separation is strongest for the most similar (most overlapping) inputs, and every neurogenesis network separates better than the control across the whole similarity range.

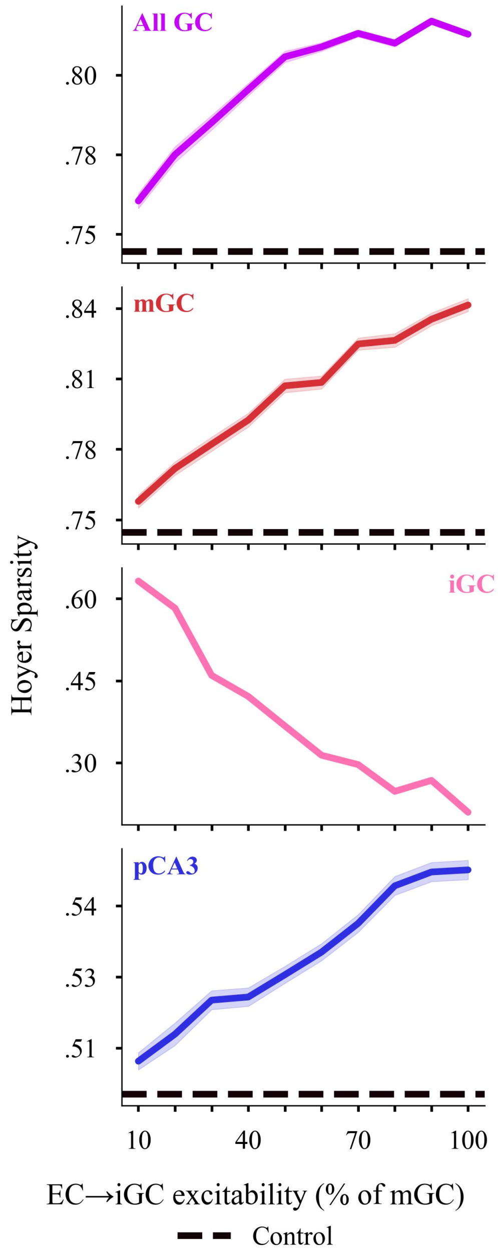

Figure S2: Population sparsity measured with the Hoyer index, as a cross-validation of the Gini results reported in the main text. Each panel shows the Hoyer sparsity of one population — the whole granule-cell population (All GC), the mature granule cells (mGC), the immature granule cells (iGC) and the pyramidal CA3 cells (pCA3) — as a function of the EC→iGC excitability level; the shaded bands show the standard error of the mean and the dashed line marks the control network without neurogenesis (absent for the iGC panel, which has no counterpart in the control). As with the Gini index, the sparsity of the All GC, mGC and pCA3 populations rises with iGC excitability and stays above the control, whereas the iGC population becomes less sparse.

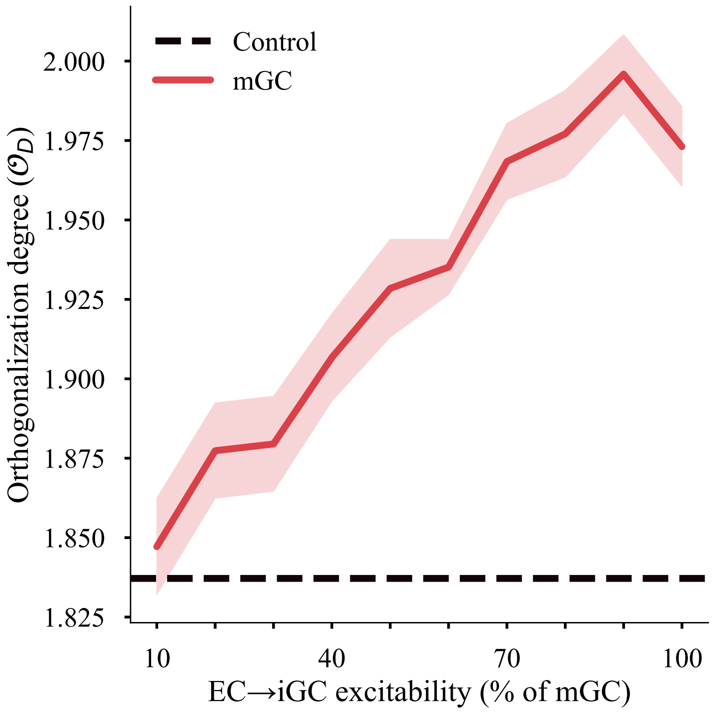

Figure S3: Orthogonalization of the mature granule-cell output as a function of EC→iGC excitability. The curve shows the orthogonalization degree $\mathcal{O}_D$ of the mGC population, taken as the ratio of the output to the input orthogonalization ($O^{(out)}/O^{(in)}$); the shaded band shows the standard error of the mean and the dashed line marks the control network without neurogenesis. The orthogonalization rises with iGC excitability, indicating that the mGC outputs become increasingly decorrelated as neurogenesis increases, beyond what the reduction in the number of active cells alone would produce.

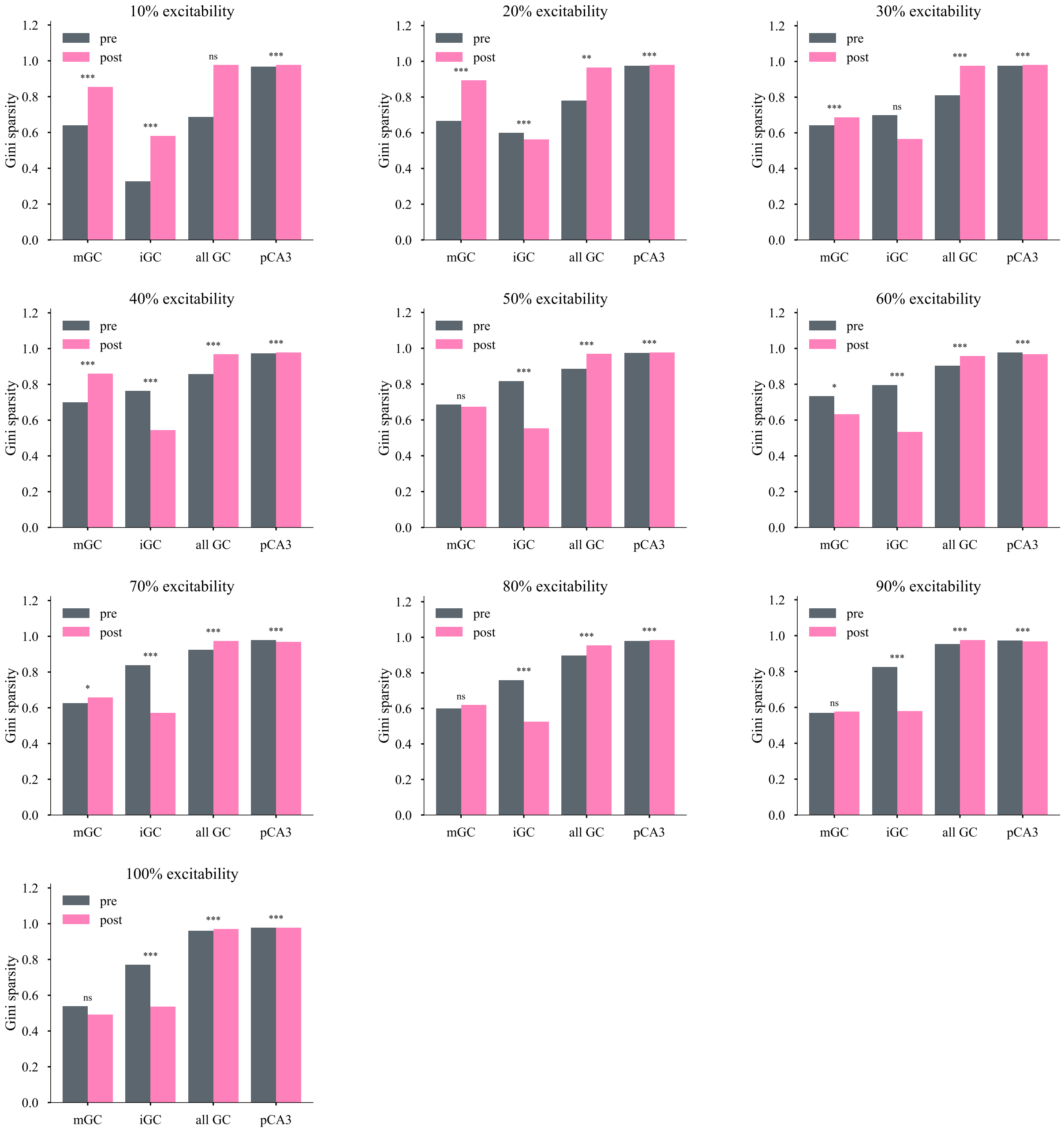

Figure S4: Population sparsity (Gini index) immediately before and after direct iGC activation, for every EC→iGC excitability level. Each panel corresponds to one excitability level (10%–100%, given in the panel title) and compares the Gini index of the mGC, iGC, combined granule-cell (all GC) and pCA3 populations in the 50 ms windows immediately before (pre) and after (post) the depolarizing pulse, pooled over all runs. Bars show the mean across runs. The result of the paired Wilcoxon signed-rank test between the pre and post values is indicated above each pair (ns, not significant; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$).

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